Phase 3I | First-coach connection development

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From named pieces to physical connections

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Preliminary engineering development. Not released for fabrication, structural testing or passenger operation.

This issue advances the Phase 3H parts-and-connections package. It develops specific alternatives for ordinary crossmember joints, seat-rail joints, deep bolster connections, the transition to shallow perimeter members, and floor/seat attachments. Dimensions and calculated demand make these options reviewable by the vehicle engineer and fabricator. They do not establish a complete vehicle or joint rating.

Development in this issue	What remains to establish
Deeper central crossmember and wider seat-rail alternatives; load and connection calculations.	Actual frame stiffness, local tube-wall behavior, effective weld lengths, combined/repeated loads, fatigue and approved welding procedures.
Bogie-reaction scenarios, bolster-to-main connection study and tapered outer bolster geometry.	Actual bogie support and sidebearer arrangement, concentrated-load reinforcement, longitudinal/lateral restraint and final interface dimensions.
Seat hardpoint reinforcement and floor-seam/bearing/retention studies.	Accepted seat base, panel edge detail, fastener/locking system, bearing and environmental qualification, and representative prototype evidence.
Dimensioned drawings, source CAD, rotatable local details and editable formula workbook.	Released manufacturing models, material certifications, tolerances, approved shop procedures and actual inspection records.

Recommendation: carry the named Phase 3I alternatives into a coordinated structural review and representative joint prototypes. Preserve the retained F1/H1 frame as the baseline until option selection, local strength, fatigue, clearance and mass consequences are accepted. None of the component catalog entries is a supplier acceptance of this train application.

01 | Continuity, datums and decision rules

The saved Phase 3H technical source archive was recovered and all 110 manifest entries were verified by SHA256 before development. It remains unchanged. Frozen copies of its frame/body inputs and design basis are included in this source package. The recovered issue is Phase 3H. It does not establish access to any separate original drawing set called Rev H; no continuity with an unrecovered set is claimed.

Basis	Retained value or rule
Coordinate convention	Inches: x increases rearward from the front inside end wall; y is transverse; z is above top of rail. Forward travel is toward decreasing x.
Frame / cabin	Frame x -47 to 353, y +/-55: 400 x 110 inches. Cabin clear 306 x 104 inches. Main girder centers y +/-34; bogie centers x45 and x261.
Study floor	F1/E1 uses a 32-inch finished floor. Top of retained steel z31.003937; nominal floor stack 25.3 mm. Earlier 28-inch baseline is preserved, not silently replaced.
Vehicle study	36-inch gauge evaluation; three regular coaches plus one accessible coach. Current work develops one regular-coach structure. Occupancy, gauge and upper masses remain engineering inputs rather than ratings.
Owner directions	0% longitudinal-grade target; maximize usable interior; farm crew performs track work within released scope; North Pole location deferred. Ride programming and business projections are outside this issue.
Current brakes	CR-G01: compressed-air service braking with spring-applied, air-released parking. Common semi-truck combination chambers remain inquiry candidates; no complete brake installation is released.

Use five evidence classes: **verified source data**, **analyst input**, **calculated result**, **preliminary selection**, and **held decision**. A trial geometry is a real defined option for review, while its actual production size, tolerance or procedure remains held where evidence is incomplete. Read the underlying input files before changing formulas.

An isolated joint model is not a substitute for the global coach model. Reactions and end moments depend on joint stiffness, support location and loading. Do not combine a fully fixed moment calculation with an unrelated simple-support capacity assumption as if both described one proven structure.

02 | Ordinary crossmembers and seat-rail end joints

PF-TRAIN-S3I-R0 • 18 October 2026 • Calculated study alternatives, not released for fabrication

What advances

Two coordinated member alternatives replace the Phase 3H oversized-fillet trials. I-CM6 deepens the 16 central ordinary crossmembers to catalog HSS 6 × 2 × 3/16. I-S32 widens all 52 seat rails to HSS 3 × 2 × 3/16, coordinating the body hardpoints. F1 and Phase 3H remain preserved. These alternatives retain the original member IDs, lengths, end planes and steel-top datum; they are not silently substituted into a released frame.

The load path remains floor or seat → crossmember or seat rail → welded endpoint → receiving HSS wall → main frame. The thicker branches and longer weld paths improve the calculated weld-metal stress. They do not establish the capacity of the receiving HSS wall or the life of the joint.

Study	Retained member	Alternative	Quantity / net length	Calculated mass change
I-CM6	HSS 4 × 2 × 1/8	HSS 6 × 2 × 3/16	16 / 1,024 in	+180.76 kg
I-S32	HSS 3 × 1 × 1/8	HSS 3 × 2 × 3/16	52 / 1,128 in	+108.30 kg

The original catalog-based frame mass is 1,527.29 kg. Applying only these two alternatives gives 1,816.35 kg. Bolster changes, hardpoint plates, welds, coatings and floor attachments are separate additions. The combination does not change the stated coach tare budget into a demonstrated mass.

Atlas catalog data are verified for these sections; stock, heat certificates and rail-vehicle suitability are not. CM6 has 0.174-inch design wall, 9.42 lb/ft, strong-axis $I = 10.5\ in^4$ and $S = 3.50\ in^3$. S32 has the same design wall, 5.59 lb/ft, $I = 1.77\ in^4$ and $S = 1.18\ in^3$. ASTM A500 Grade C remains a material candidate requiring the project's toughness and dynamic-service review. [I-F01]

I-CM6 dimensions and calculations

The worked T93_1 branch occupies x = 92-94, y = −32-32 and z = 25.00394-31.00394 inches. Its 64-inch end-plane separation and 2-inch x width are unchanged. The new bottom is 2 inches lower. G1 remains HSS 12 × 4 × 1/4 at y = 32-36. Brake, air-line and bogie swept clearances must be checked against this deeper member.

Two trial 3/16-inch vertical fillets lie at x = 92 and 94 on the y = 32 connection plane. Each drawn root path is 4.5 inches, from z = 25.75394 to 30.25394. The 3/4-inch top and bottom setbacks put the root paths inside both members' catalog vertical workable flats. They do not prove actual corner shape, root gap or torch access. No weld is added above the floor-bearing plane.

The retained floor sensitivity uses 4.8 kPa live plus 1.0 kPa assumed dead pressure, a 24-inch tributary width, actual catalog member self-weight and a factor of 2 on total gravity. It is an analyst sensitivity, not an adopted rail load factor. Seat occupants are not added again to the same floor-live case. The branch sees 5.971 kN end reaction and 1.618 kN·m fully restrained end moment. The receiver's 2-inch face-to-axis offset also produces 0.303 kN·m about the main-girder axis; the local connection screen does not qualify that receiver torsion.

The elastic two-line weld model uses $a = leg/\sqrt{2}$, $A = 2aLe$ and $I = 2aLe^3/12$. Resultant stress combines direct shear and end-moment normal stress. The reference weld-metal comparison is 0.75 × 0.60 × 70 ksi = 217.18 MPa, without directional enhancement. It is not a vehicle connection acceptance value. [I-F02, I-F03]

Effective length assumed per side	Fully restrained resultant	Weld-metal comparison ratio
4.500 in, 100% of drawn length	110.58 MPa	0.509
3.375 in, 75%	196.38 MPa	0.904
2.250 in, 50%	441.52 MPa	2.033

The half-length case exceeds the reference comparison. Actual effective length is therefore a decisive engineering input, not a drafting convention. The new study's nominal weld leg equals nominal branch wall, but still exceeds the 0.174-inch design wall. Minimum and maximum weld size, cut-edge conditions, burn-through, undercut, process and WPS remain unaccepted.

The retained 4 × 2 × 1/8 member with the Phase 3H 1/4-inch × 3-inch fillet trial has a ratio of 0.842 under its own self-weight. A 4 × 2 × 1/4 comparison with two 1/4-inch × 2.5-inch drawn weld paths gives 1.232. Simply adding wall thickness does not rescue that particular shorter two-line detail. This is not a rejection of every possible 4-inch-deep joint.

The calculated membrane-strip stresses are 84.68 MPa in the CM6 branch and 63.23 MPa in the main wall at full assumed effectiveness. Those numbers do not include receiver-face flexure or punching. The main's connecting face is 12 inches vertically, with a top-aligned 6 × 2-inch branch patch; treating the chord-face width as 4 inches or assuming a centered truss joint would be wrong. [I-F04]

I-S32 endpoints and horizontal seat forces

All 64 retained H-S16 hardpoints are mapped onto the actual 21- or 22-inch rail segments. The 104 endpoints comprise 44 receiving CM6 central crossmembers, 44 receiving retained 4 × 2 × 1/8 outer crossmembers, and 16 receiving deep bolsters. This phase does not describe every receiver as strengthened.

The calculation enumerates six separate cases: gravity; 1.5 times vertical gravity; both signs of 1g longitudinal force; and both signs of 0.5g lateral force. Simple and fully restrained endpoint limits produce 1,248 demand records. These are analyst sensitivity cases using 305 kg per occupied bench, a 24-inch CG above the floor and the 16 × 24-inch foot pattern. They are not a seat certification or crash-load specification.

Longitudinal overturning gives 2.991 kN compression and 1.496 kN uplift at individual feet. Horizontal forces are carried into the rail, not ignored. Because actual longitudinal restraint sharing is unknown, each endpoint is separately bounded by the entire net horizontal force on its rail. The 2.496-inch floor-to-rail-center offset contributes additional moment. The model adds adverse moment magnitudes; it is a conservative endpoint envelope, not a simultaneous balanced frame solution. Back-rest loads, concurrent acceleration axes and abuse are still open.

The controlling drawn example is S-44_10 at its x = 238-inch start face against T237_0. The rail ends at x = 259, is centered at y = −44 and spans y = −45 to −43. Two hardpoints are at x = 241 and 257. Its side welds are trial 3/16-inch fillets, 2 inches long, at y = −45/−43 from z = 28.50394 to 30.50394.

For the governing longitudinal case the bounded endpoint demand is 2.809 kN vertical, 1.496 kN axial and 0.258 kN·m bending. The three weld-metal ratios for 100%, 75% and 50% assumed effective length are 0.431, 0.757 and 1.680. The wider rail does not remove the thin receiver: T237_0 still has 0.116-inch design wall. Its membrane-strip screen is 107.88 MPa, but the receiving-face local resistance remains unknown.

The combined weld-group calculation includes axial normal stress, two bending axes, two shear components and torsion about the rail axis. The JSON retains all inputs, footprints, forces and resulting cases so a reviewer can change the load and restraint model.

Connections still needing engineering closure

Receiver-face plastification, punching, nonuniform effective width, local buckling, chord utilization and fatigue require appropriate local connection analysis. A generic added patch plate is not credited: the plate type must match the actual governing mechanism and tension/compression behavior. [I-F04, I-F05] The finite-element route, if used, needs verified geometry and boundary conditions, mesh and material checks, and comparison with known behavior. No finite-element capacity is claimed here. [I-F07]

Static weld stress does not establish service life. The route's joint impacts, wheel loads, body vibration, braking/reversing spectrum and number of cycles must feed an accepted fatigue method. Weld root/toe behavior and start/stop locations require deliberate treatment. No fatigue category from a pole or building detail is transplanted to this train. [I-F06]

Before a trial joint is fabricated, obtain measured section corners and wall/seam locations, accepted materials and toughness, global load combinations, effective weld lengths, local wall checks, WPS/PQR and welder requirements, access/root-gap limits, dimensional tolerances and inspection acceptance. The drawings show study dimensions; actual saw cuts, welding release and approved joint capacities remain null.

03 | Bolster load path and perimeter transition

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The useful advance is a physically defined tapered outer bolster and a measured, inspectable central-joint option. The tapered box removes the previous eight-inch depth mismatch at the shallow perimeter rail. The central-joint calculation also identifies a limitation: the twin-angle weld trial cannot carry the larger root moments in this sensitivity study. The angles therefore remain an option to evaluate, not a selected support connection. Actual bogie support geometry and complete frame stiffness must be resolved before this primary load path can be released.

Retained geometry and explicit changes

Keep F1's main HSS 12 × 4 × 1/4 girders at y = ±34in, central deep bolster segments at x = 45 and 261in, 64in between main inner faces, perimeter HSS 4 × 2 × 1/8 at y = ±54in, and steel top z = 31.003937in. The separate F1 floor height remains a 32in study. No main rail is cut, slotted or silently replaced. The original four 17in outer bolster HSS pieces remain in Phase 3H; I-BT is a replacement alternative with explicit IDs.

I-BA adds two exterior vertical L3 × 3 × 1/4 angle trials at each central bolster end, eight angles total. Each is 10.5in tall, z = 19.753937 to 30.253937in. This height fits inside the retained G12 catalog 10.875in workable flat depth, but catalog dimensions do not establish actual corner radii or close fit. The CAD angles use sharp ideal corners; measured heel/toe radii, weld access and actual root contact must be checked on a sample. There are no bolt holes or improvised bogie fasteners.

I-BT replaces T45_0, T45_2, T261_0 and T261_2. Each four-plate box occupies the existing x-width of 4in and the existing 17in transverse length. Its top stays flush. Its depth tapers from 12in at y = ±36in to 4in at y = ±53in. Thus the outer end matches the nominal perimeter depth instead of leaving eight inches of deep member hanging below its receiver. Matching depth alone does not qualify the joint into the 1/8in-wall perimeter.

Loads and support sensitivity

The 13,200kg upper coach allocation is conservatively treated as body-supported mass, without subtracting bogie or other unsprung mass. Actual measured tare, passenger distribution and equipment positions have not been established. Vertical multipliers of 1.0, 1.5 and 2.0 are analyst sensitivity cases, not adopted regulatory load factors. Bogie load shares of 50% and 60%, and transverse main-rail shares of 50% and 60%, expose load-distribution sensitivity.

For coach mass m, vertical multiplier k, bogie share f and main share q, the studied reactions are P = m g k f and V = P q. At k = 2 and f = q = 0.5, one bogie carries 129.448kN and each main carries 64.724kN. At f = q = 0.6, the larger main reaction is 93.202kN. A 60/40 transverse split needs a roll-support couple P(2q − 1)(34in), or an appropriately shifted bearing reaction. A freely rotating center point cannot resist that couple by itself. For P = 155.337kN, that couple is 26.830kNm. Side-bearer gaps, contact stiffness and the resulting load distribution are unresolved.

A separate symmetric transverse-beam comparison uses downward main loads at y = ±34in. With a center point, peak moment is P(34in)/2. A centered, uniformly loaded support patch of transverse width b gives peak moment P[34in/2 − b/8]. Equal support points at y = ±s give P(34in − s)/2. These are idealized support probes, not dimensions for a center bearing or side bearers.

Symmetric 2g, 50% bogie share	Peak transverse moment	Gross elastic stress using S = 19.8in³
Center point	55.90kNm	172.27MPa
Centered 12in transverse patch	50.96kNm	157.07MPa
Equal supports at ±18in	26.30kNm	81.07MPa
Equal supports at ±24in	16.44kNm	50.67MPa
Equal supports at ±30in	6.58kNm	20.27MPa

These figures are gross beam screens only. They omit local bottom-wall bending, bearing/crushing, longitudinal footprint, diaphragm load distribution, torsion, buckling, connection compliance, impact and fatigue. A nominal support directly beneath each main makes this ideal beam's transverse moment zero; it does not make local stresses zero. Gauge alone does not establish room for any of these support locations.

Before selecting a central-joint detail, obtain the engineered bogie's mounting arrangement, vertical and horizontal reactions, uplift retention, roll and yaw freedom, bearing footprint, permissible contact pressure, wear allowance and brake-hose swept envelope. Develop the center-bearing/load-spreading diaphragm or other approved support arrangement together with a frame stiffness model. The four-inch-wide central HSS underside must not be treated as an accepted bearing pad.

Twin-angle joint and weld demand

At the front right central joint, the main inner face is y = 32in and the bolster side faces are x = 43 and 47in. One angle sits in each empty outside corner. The two main-side weld toes are x = 40 and 50in, y = 32in; the two bolster-side toes are x = 43 and 47in, y = 29in. Each vertical weld study runs from z = 20.253937 to 29.753937in: a 9.5in length with a 3/16in fillet leg. Those exact line locations and mirrored parts appear in bolster_i.json. Original direct end welds receive no shared-load credit in this screen.

The load path is bolster sidewalls → two angle-side toe welds → angle legs → two main-side toe welds → main face → the rest of the main section. Both interfaces must work. Adding angles does not remove local main-face flexibility or transfer the load magically into the far main wall.

For either two-line weld group, a = weld leg/√2, A = 2aL and I = 2aL³/12. The screen combines V/A with M(L/2)/I by root-sum-square. M includes V × 3in for the offset between toe attachment lines, plus a root moment sensitivity. For a symmetric 64in beam with fully fixed ends and a center point load, fixed-end moment is P L/8 = V × 16in. Multipliers α = 0, 0.25, 0.5 and 1.0 probe that magnitude. They do not establish actual joint rotational restraint; using the same scale with a 60/40 transverse split is not an exact asymmetric frame solution.

The reference stress is 0.75 × 0.60 × 70ksi = 217.185MPa, with no directional enhancement. It is a static weld-only comparison derived from a general structural reference, not an adopted rolling-stock design code or approved allowable.

2g reaction case	α = 0	α = 0.25	α = 0.5	α = 1.0
50% bogie / 50% main	0.393	0.831	1.287	2.208
60% bogie / 60% main	0.566	1.197	1.853	3.180

Values are weld stress/reference ratios. Ratios above 1 exceed this limited comparison. In the 60/60 case, the weld-only remaining root-moment threshold is 6.589kNm, corresponding to α = 0.174, after allowing for the 3in eccentricity. The balanced threshold is 9.023kNm, α = 0.343. These are mathematical bounds for this weld model, **not connection moment ratings**. The gross bolster-side angle-leg in-plane bending screen also reaches 298.8MPa in the largest sensitivity, above the candidate A36 minimum yield stress of 248.2MPa. Main-side leg bending, angle torsion and prying are not resolved by that in-plane screen. A favorable shear-only weld result cannot select this detail.

Connection completion must evaluate effective weld length, local chord/branch stresses, angle heel/toe geometry, combined force and moment, actual load sharing, fatigue stress ranges and the accessible welding/inspection procedure. More weld metal alone will not solve an unfavorable frame load path.

Tapered box plates and screening

Each I-BT box uses candidate 1/4in ASTM A572 Grade 50 plate. Two trapezoidal webs fit between a full-width top and sloping bottom plate, with no solid overlap. The web root height is 11.473702in and the tip height 3.473702in. Top plate finished envelope is 17 × 4 × 1/4in. The bottom plate has 1/4in **normal** thickness; at its 25.201° slope this occupies 0.276298in vertically. Its centerline sloped length is 18.788294in, with vertical end cuts. Exact global polygons control; decimal values are idealized geometry, not approved manufacturing tolerances.

The four flush longitudinal corner seams need a designed edge/groove weld detail, effective throat, preparation, backing/access sequence and inspection plan. They are not four generic fillet welds. Seam welds and both end welds remain null. The shown plates are finished study envelopes without bevels, root gaps, trim allowances or weld shrinkage compensation; they are not released cutting blanks.

Independent 1, 5 and 10kN downward loads at the outer tip assess section stiffness. These are explicit unit-load sensitivities, not resolved roof/wall/passenger reactions. At 10kN, root moment is 4.318kNm. A 101-station sweep of the actual ideal box sections finds 11.829MPa maximum gross bending stress at 1.53in from the root and 8.924MPa maximum average web shear. Virtual-work integration gives 0.0435mm bending-only deflection. Shear deformation, local plate behavior, taper effects, end flexibility, torsion and fatigue are not included. This suggests the gross tapered section is not the immediate issue under that load probe; end attachment to the thin perimeter and complete load combinations still need engineering.

The four boxes weigh 51.103kg by ideal plate volume, replacing 66.367kg of catalog HSS. Eight sharp-angle envelopes add 15.533kg. Combined nominal change is +0.269kg before weld metal, later reinforcements, hardware and actual rolled radii. No material order or supplier quotation has been obtained.

Fabrication development and release sequence

1. Review the support interfaces and frame model with the bogie supplier and structural engineer; settle center-bearing/side-bearer load distribution and actual combination loads.

2. Select the main/bolster local reinforcement, joint stiffness and cyclic design basis. Recalculate the angle option or replace it if its demand bound is insufficient.

3. Have the fabricator review a full-size angle/box corner sample for flat contact, welding access, distortion control and inspection access. Do not fabricate the coach's primary support joints from this revision.

4. Complete tapered-box seam and end details, perimeter reinforcement, material toughness requirements, WPS and acceptance criteria. Keep interior corrosion protection, drainage and sealing compatible with inspection.

5. Release controlled shop drawings only after calculations, drawings and first-article test requirements agree. Farm staff can help with dimensional mockups and handling plans; qualified specialists must own the support weldments, inspection and vehicle release.

Source notes

I-B01-I-B03 explain why connection fixity, effective HSS walls and local limit states need explicit treatment. I-B04 supplies the candidate A572 Grade 50 material minima. I-B05 is a real 3 × 3 × 1/4 angle listing, but its A36 title and ASTM-A529 standard field conflict: obtain an actual mill certificate before accepting a grade. I-B06 concerns cyclic stress and life; no fatigue category or life has been adopted here. I-B07 retains Atlas G12 section properties. I-B08 supports the static weld comparison only. Full links, verification dates and limitations are in bolster_i_inputs.json and bolster_i.json.

04 | Floor seams and seat hardpoints

Connection-development revision, 18 October 2026. Not for fabrication. This module turns the previous mounting markers into dimensioned steel and floor-interface trials. It preserves the 32-inch F1 floor study, H-S16 foot centers and H-P125 panel envelopes. It does not establish a production seat, qualified panel assembly or approved attachment capacity.

I-S32: a wider seat rail

The separate I-S32 option replaces all 52 HSS 3 × 1 × 1/8-inch seat rails with **HSS 3 × 2 × 3/16**, retaining their centerlines, 3-inch depth, top plane and face-to-face lengths: 36 at 22 inches and 16 at 21 inches. Atlas publishes 5.59 lb/ft, 1.77 in⁴ strong-axis inertia, 1.18 in³ section modulus and a 0.174-inch design wall for the new section. Its 1.156-inch catalog top flat is still not permission to drill an unsupported fastener hole. [I-D01]

The increased width allows bolts and nuts outside the closed tube. Seat loads pass through the soleplate, steel spacers, cap and reinforced side attachments into the seat rail; the sandwich floor is bypassed. The rail-to-crossmember joints are coordinated in the frame module. Actual rail material toughness, local wall behavior, load sharing and fatigue remain unverified.

I-SH1: a removable, accessible hardpoint

Each retained foot center receives the following candidate parts. Dimensions describe the revised study, not production cut tolerances.

Part	Quantity per foot	Candidate dimensions
Cap	1	A36, 2.50 × 6.00 × 0.375 in
Side cheek	2	A36, 2.50 × 2.00 × 0.1875 in
Seat soleplate	1	A36, 4.25 × 6.50 × 0.250 in
Rigid spacer	2	1.000 in OD, 0.5625 in bore, 15.775 mm nominal length
Bolt assembly	2	1/2-13 UNC SAE J429 Grade 5 candidate; complete assembly held
Plate holes	2 per cap/sole	9/16 in, on a 4-inch transverse pitch

A36 finished plate has a published minimum yield strength of 36 ksi; it must arrive with appropriate certification. The spacer requires a certified machinable blank with accepted properties; unspecified pipe is not interchangeable. [I-D02] The Grade 5 specification comparison gives 85 ksi proof stress for these diameters, while the published 1/2-13 tensile area is 0.1419 in². These are material and thread data, not a seat-attachment rating. [I-D03, I-D04]

The representative foot is at **x97, y20, z32 inches**. Bolt axes are x97/y18 and x97/y22. Steel top is z31.003937; the cap reaches z31.378937, the spacers reach z32 and the soleplate reaches z32.25. The nominal spacer length is therefore **25.30 − 9.525 = 15.775 mm**. Measure the accepted stack before releasing spacer machining. Coatings, flatness and any final isolation layer must be included; a loose shim stack is not defined here.

The cap and cheeks were shortened from the early 3.75-inch trial after checking every foot against the rail-end welds. All 64 final caps have at least 0.75 inch to a transverse member face and 0.5625 inch after reserving the proposed 3/16-inch rail-end fillet projection. Each bolt has a 1.25-inch-diameter tool-cylinder reservation below the cap, with 0.1875 inch to the cheek. Nominal tool cylinders do not intersect the surrounding frame. These are geometric checks; actual sockets, washer/nut thickness, hand access and welding access still require a physical mockup.

A **2.75 × 6.25-inch finished, closed opening** surrounds each hardpoint in its panel. Every opening lies within one retained panel; minimum clearance to a finished panel edge is 2.5625 inches. The soleplate overlaps that opening by 0.75 inch longitudinally and 0.125 inch transversely. The overlap is a sealing interface, not a floor bearing surface. OEM reinforcement and a maintainable waterproof closure must be developed; do not field-cut raw honeycomb from these envelopes. The soleplate stands 1/4 inch above the floor and projects 1/8 inch beyond the nominal bench-depth envelope at the outer feet. The eventual seat must cover/protect this detail and adjust pedestal height to retain the 18-inch seat-height reference.

Sizing evidence and its limits

Retained analyst loads use two 145 kg adults and a 15 kg bench allowance. The separate longitudinal 1g case at a 24-inch combined CG gives, under a rigid four-foot assumption, **2.991 kN compression, 1.496 kN uplift and 0.748 kN horizontal shear at a foot**. These are sensitivities, not adopted passenger-restraint criteria. Adding this hardware does not certify the earlier bench-mass allowance or CG.

For a plate strip, the elastic screen is $\sigma = 6M/(bt^2)$. The cap-wing model uses a 1-inch lever from the rail side to the bolt axis. The soleplate model uses a central force over the 4-inch spacer span. At full assumed strip width, cap stress is 39.56 MPa with equal bolt sharing, 79.12 MPa with the entire foot force on one wing, and sole stress is 104.72 MPa. Halving effective widths raises the one-wing cap result to **158.25 MPa** and the sole result to **209.44 MPa**, versus the 248.21 MPa material-yield comparison. Sole deflection becomes 0.284 mm in that reduced-width sensitivity. The rejected 1/4-inch cap reached **356.05 MPa** in the combined half-width/one-wing case, which drove the 3/8-inch trial selection. No safety factor, fatigue life or accepted plate capacity is implied by comparison to yield.

Equal sharing between two bolts gives 8.17 MPa nominal tensile stress. Assigning all foot uplift to one bolt and multiplying by two for an additional prying sensitivity gives 32.67 MPa; that multiplier is not a demonstrated bound on prying. Bolt shear, bending, real soleplate stiffness, pretension and vibration retention need a complete assembly analysis. A separate 70%-of-proof preload sensitivity is 37.56 kN per bolt; including service compression gives 112.74 MPa average spacer compression. **That is not a specified preload or torque.** Washer/spacer contact, local plate bending from clamping and thread engagement remain held.

Each cheek has two 1/8-inch trial vertical fillets, each 1.5 inches long, on the tube side flat. For longitudinal loading, the four-line elastic group is treated in its x-z plane: $\tau x = V/A + My \cdot z/J$; $\tau z = -P/A - My \cdot x/J$. The maximum corner result is **12.98 MPa**. The cap-to-cheek 3/16-inch longitudinal fillets have only a direct-force screen; their moment/prying capacity remains unresolved. The 217.18 MPa weld reference derives from 0.75 × 0.60 × 70 ksi and is a comparison, not the vehicle welding basis. [I-D07] Local tube-face behavior, lateral seat demand, effective weld length, residual stress, fatigue and qualified procedures govern further development.

I-FB / I-FR: bearing and retention separated

At the representative ordinary crossmember seam x93, retain the **1/8-inch panel gap**. Two candidate rigid glass-epoxy strips, each **7/16 inch wide × 3 mm thick**, flank the gap. Their outer edges have 0.078-inch nominal margin to the new CM6 catalog flat. Under the retained 4.8 kPa floor comparison load and 24-inch support spacing, reaction is 1.463 kN/m per edge and average strip pressure is **0.1317 MPa**. This excludes caster effects and imperfect bearing.

Norplex NP510A is an identifiable glass-epoxy material with typical flatwise compression of 455.1 MPa; the manufacturer explicitly distinguishes typical data from specification values. [I-D05] No allowable is assigned. The 3 mm machined thickness, long-term compression, uneven contact, moisture, fire/smoke behavior and substrate compatibility need qualification. A 1-inch closed-edge envelope is reserved on each panel, but its internal construction is held. Plascore offers factory closeouts and inserts; no suitable project rating has yet been provided. [I-D06]

The underside retention prototype uses paired two-piece A36 clips, one on each side of the crossmember. Each horizontal flange is **2 × 1.5 × 3/16 inches**, with a 7/16-inch study hole for a 3/8-16 candidate bolt. The bolt is 3/4 inch outboard of the HSS face and must engage an OEM structural insert, never the 0.020-inch skin alone. Its axis is 1.6875 inches from the finished panel edge, beyond the 1-inch edge-closeout reservation; it therefore needs a separate supplier-designed hardpoint or extended

reinforcement. The vertical web is 1.5 inches wide, 3/16 thick and 2.180610 inches high at the nominal unisolated interface. Its top must be lowered by the accepted electrical-isolation/barrier thickness. A positive closure and corrosion detail remain to be designed.

A **1 kPa uplift and separate 1 kPa in-plane sensitivity over a 24 × 24-inch patch** give 0.372 kN in each direction at a clip. Flange bending is 49.15 MPa. Combining force and eccentric moments at the two vertical frame-weld lines gives a 20.69 MPa peak comparison. These results do not qualify the flange/web elbow weld, insert pullout, panel closeout or attachment fatigue. Production spacing, total clip quantity and controlled thermal movement are deliberately held; the illustrated pair is not a complete floor fastening plan.

Mass and next evidence

The wider rails add **108.30 kg**. The 64 six-piece hardpoint sets add **121.54 kg**, giving **229.84 kg** before fasteners, welds, floor clips and panel reinforcement. Count the rail increase only once if included in the frame revision. These maintainable attachments are a comparison option, not the lightest accepted design.

Next obtain a real seat/base assembly and supplier-designed panel closeouts/inserts, establish the governing load spectrum, and complete the local HSS, plate/prying and weld assessments. Build an engineer-approved first article with measured tool access, stack, corrosion isolation and sealing. Release production dimensions and inspection criteria only after that evidence. The farm crew can make unloaded templates; loaded attachment tests and safety-critical fabrication require the responsible engineering and fabrication specialists.

Coordinated mass comparison

Listed study contribution	Mass
Retained F1/H1 steel frame	1,527.29 kg
16 deeper central crossmembers, net change	+180.76 kg
52 wider seat rails, net change; counted once	+108.30 kg
Four tapered outer boxes and eight angles, net ideal change	+0.27 kg
64 machined hardpoint steel sets	+121.54 kg
Frame plus these listed alternatives	1,938.15 kg

The listed alternatives add 410.87 kg to the retained steel frame. This is an additive comparison, not a selected complete frame or measured coach tare. Do not add the body subtotal again: it already includes the wider rails. Excluded: Bolts, nuts, washers, welds, coatings, floor retaining system, closed edges, bogie support hardware, local reinforcement still to be designed and complete body/equipment.

The external seat hardpoint is deliberately explicit and maintainable, but its material and machining burden merits further optimization after the actual seat, loads and fatigue basis are accepted. A lighter option must still carry the same accepted forces and permit assembly, tightening and inspection.

05 | Fabrication development and inspection records

The following sequence is an engineering and first-article plan. A start condition must be met before the corresponding structural work proceeds. Actual tolerances, weld procedures, test limits, measured results and signatures are intentionally blank in the editable source records.

I-QA01 | Choose the coordinated alternative

Responsible: Responsible vehicle structural engineer.

Start condition: Issued option selection and whole-frame load path; accepted vertical, lateral, braking, reversing, coupling, twist, fatigue and lifting cases.

Record: Option IDs; signed load-case table; drawings replaced; mass/CG delta; actual selection.

I-QA02 | Obtain material and component evidence

Responsible: Purchaser with engineer and fabricator.

Start condition: Approved material specification, toughness requirement, mill heat/certification, actual dimensions and source. Catalog listings alone do not confirm stock or vehicle suitability.

Record: Part ID; heat/lot; mill certificate; section wall/corner readings; supplier lead time; accepted substitution.

I-QA03 | Qualify each welded joint family

Responsible: Welding engineer and qualified fabricator.

Start condition: Joint design, WPS/PQR as required by adopted basis, personnel qualifications and access sequence. Include root gap/preparation, fillet or groove extent, starts/stops, heat input, distortion and repair procedure.

Record: Joint ID; drawing revision; WPS and qualification; fit-up values; welder; inspection method and acceptance.

I-QA04 | Build a sacrificial joint coupon

Responsible: Qualified shop with structural test engineer.

Start condition: Approved coupon plan reproducing actual tube wall/corners, mating plates, weld position and load eccentricity. Fixture and test capacities must exceed the accepted test loads.

Record: As-built dimensions; material heat; weld procedure; photographs; calibrated load/displacement; failure mode.

I-QA05 | Check floor and hardpoint prototype

Responsible: Panel supplier, seat supplier and structural test engineer.

Start condition: Qualified panel edge/insert/bearing details and representative seat base. Test load directions and combinations, retention, fastener locking, finish and water ingress before occupied trials.

Record: Actual panel/core/skin; edge closeout; bolt grip/torque; spacer fit; bearing contact; permanent set; observed damage.

I-QA06 | Set and measure the frame fixture

Responsible: Fabricator with engineer; trained farm crew may assist.

Start condition: Released datum setup and tolerances. Fixture supports must not force parts into the desired shape.

Record: Datum A top plane; B centerline; C front face; bolster x45/261; girder y±34; diagonals; twist and straightness.

I-QA07 | Inspect before closing or coating

Responsible: Designated inspector.

Start condition: Accepted inspection plan, visual access to all required welds and any NDE required by the adopted joint class. Resolve defects through controlled repairs.

Record: Joint ID; method; actual result; inspector; nonconformance; repair reference; reinspection.

I-QA08 | Install floor and seats

Responsible: Qualified installers; farm crew only within released scope.

Start condition: Accepted structure, insulation/bearing stack and fastener installation instructions. Maintain access to nuts, drains and inspection faces.

Record: Part IDs; contact fit; isolation continuity; fastener lot/grip/torque and witness mark; drainage; final floor elevation.

I-QA09 | Release the first structure to vehicle integration

Responsible: Responsible engineer and inspector.

Start condition: Accepted first-article dimensional and structural evidence; actual weighed mass/CG and all open exceptions reconciled. This is not passenger commissioning.

Record: Issue/release signatures; as-built drawings; mass/CG; outstanding limits; handoff to running-gear/brake integration.

The farm crew can organize labels, templates, parts, photos and measurement records and assist with handling under an issued procedure. Structural fabrication, welded joints, bolted seat anchors, bogie interfaces and testing require the qualifications and supervision assigned in the released work packages. This issue does not authorize a structural build from the study drawings.

For the welding basis, AWS currently lists D15.1/D15.1M:2025 for railroad cars, locomotives and components, including provisions for thin base metal. Its public preview verifies scope and contents, not detailed compliance. The responsible engineer must decide the applicable project specification and acquire the adopted edition and amendments. [I-Q01]

Reinforcement must address the governing limit state, not simply add metal. Published HSS guidance treats defined connection geometries and distinguishes their failure modes. These coach details require their own applicability check. [I-Q02]

If local finite-element analysis is used, require realistic corner geometry, wall thickness, weld/load transfer, support stiffness, material behavior, mesh convergence and validation. A rendered or tessellated CAD model is not structural FE analysis; none is claimed in this issue. [I-Q03]

06 | Procurement and engineering handoff

Package	Evidence to obtain before ordering or fabrication
Structural tubes and plates	Exact size/specification; mill availability, length and lead time; certified heat and toughness requirements; actual wall/corner geometry; substitution process. Listed catalog sizes do not establish local stock.
Seat bases and fasteners	Accepted seat/base drawing and interface loads; exact bolt/nut/washer grade and coating; thread location, grip length, tightening/locking procedure and access. Fastener tensile data alone is not attachment capacity.
Floor sandwich panels	Actual core/skins/adhesive; finished thickness/tolerances; closed-edge/insert detail; supported span, concentrated/caster loads, fastener pull-through and environmental qualification.
Bogie assemblies	Complete support/contact and retention drawings; rated load and duty; suspension/sidebearer travel and forces; wheel/rail geometry; brake envelope; weld/bolt restrictions on supplied parts.
Fabrication and inspection	Adopted design/welding basis; accepted joint drawings and WPS/qualifications; fixture plan; tolerances; inspection access/methods; repair acceptance and first-article documentation.

No supplier has been contacted, no stock or lead time has been confirmed, and no purchase order has been placed in this phase. Links are research and inquiry references. The public site remains a technical review edition; commercial budgets and demand projections are excluded.

Next development should integrate the selected connection options with the actual running gear, brake and body interfaces, then complete the global/local/fatigue verification and representative joint/seat-floor prototype plans. Until the supporting evidence is accepted, a more complete-looking production drawing would imply a precision the design has not established.

07 | Deliverables and verification limits

File group	Purpose and limits
Dimensioned PDF/SVG/DXF sheets	Named details, datums, dimensions, material candidates, sections and assembly interfaces. See the drawing README for paper-inch versus actual-inch DXFs. Dimensions govern; do not scale a page image.
STEP and parametric source	Local connection assemblies and exploded views; STEP millimetres, canonical coordinates/browser meshes inches. STEP is an exchange solid, not a native feature history. Source scripts retain editable parameters.
Connection workbook	Linked calculation inputs/results, option mass changes, sources and held decisions. Formula and cache checks are numerical coordination checks, not professional structural acceptance.
Source package	Canonical JSON, calculation scripts, revision/source registers, drawings/models, workbook source and review records. Frozen predecessor inputs permit traceable reruns.
Technical site	In-browser report and drawing readers plus local assembled/exploded detail viewer. Prior issues and the separate pivoting-bogie demonstration remain available.

Verification includes source-to-drawing/model coordination, calculation arithmetic, file integrity, formula/cache checks, and visual review of the PDFs. These checks do not establish physical strength, fatigue life, weld quality, manufacturing fit, braking performance, vehicle stability or regulatory approval. No real joint, seat, floor or vehicle test has been performed.

A separate calculation review identified and corrected the seat-cheek weld force-direction treatment, checked 96 bench/load equilibrium groups, reconciled the wider rails once in the mass comparison, and checked final attachment dimensions against the model sources. The review record states its bounded scope and does not constitute professional engineering approval.

A supported interactive browser was unavailable in this session. Site code, references and data were checked locally; actual browser rendering and interaction have not been verified here. Static drawings and model views provide an additional reading path.

08 | Dated source register

Accessed 18 October 2026 unless an entry says otherwise. Internal predecessor records are identified separately from external primary evidence. Source facts, calculated results and author-selected geometry must not be interchanged.

I-F01 / I-B07 / I-D01 | Atlas Tube ASTM A500 rectangular HSS section properties

https://www.atlastube.com/wp-content/uploads/2018/04/Atlas-HSS-Section-Properties-A500-Rectangles.pdf

Page 5: HSS6x2x3/16 and HSS4x2x1/4 dimensions, catalog mass, design wall, strong-axis properties and workable flats; retained rows checked in Phase3H. Retained Phase 3H G12 catalog properties S 19.8in3/I 119in4/design wall 0.233in/mass 25.82lb-ft; original Atlas URL re-opened this phase. Page6 HSS3x2x3/16: 5.59lb/ft,A1.54in2,I1.77in4,S1.18in3,designwall.174in,flatwidth1.156in.

Limit: Published geometry/property data are not stock availability or a material certificate. Catalog nominal properties, not supplier stock, actualcornergeometry, heat certification or passenger-vehicle approval.

I-F02 | Steel Tube Institute - Design of Fillet Welds to Rectangular HSS

https://steeltubeinstitute.org/resources/design-of-fillet-welds-to-rectangular-hss/

Effective weld length and local weld eccentricity matter in rectangular HSS joints; no directional strength enhancement is used.

Limit: Article is technical reference, not project welding acceptance.

I-F03 / I-B08 / I-D07 | Steel Tube Institute - Full-Capacity Welds

https://steeltubeinstitute.org/resources/full-capacity-welds-cjp-vs-fillet-vs-fillet-reinforced-pjp/

Reference comparison uses phi 0.75 and nominal weld stress 0.60 FEXX; base-wall yield comparison uses phi 0.9. Reference phi0.75 and nominal weld strength0.60FEXX; no directional enhancement. Static comparison only, not selected rail welding/fatigue basis. Fillet weld comparison phi.75,nominal.60FEXX.

Limit: No full-capacity weld or vehicle joint is established by this screen. Not adopted rollingstock welding basis; no fatigue/localHSS/fit-up/WPS release.

I-F04 | Steel Tube Institute - HSS Limit State: Plastification

https://steeltubeinstitute.org/resources/hss-limit-state-plastification/

HSS connecting-face out-of-plane yielding and punching are distinct local limit states.

Limit: A through-thickness strip stress is not the face flexural resistance.

I-F05 / I-B02 / I-Q02 | Steel Tube Institute - Plate-Reinforced HSS Connections

https://steeltubeinstitute.org/resources/plate-reinforced-hss-connections/

Reinforcement type must address the governing failure mode; chord-face plates and sidewall plates have different functions. External reinforcing plates do not eliminate chord/branch limit states; applicability and connection behavior matter. Reinforcement is tied to particular connection configurations and limit states; a plate is not an automatic increase in complete joint capacity.

Limit: No generic capacity assigned to an uncalculated patch plate. Article develops specified static geometries. This coach has different combined/repeated loads; do not transfer formulas without confirming applicability.

I-F06 | Steel Tube Institute - HSS Poles

https://steeltubeinstitute.org/resources/hss-poles/

Cyclic weld detail and start/stop position influence fatigue; static strength alone does not establish fatigue life.

Limit: Pole detail fatigue category is not transferred to this train frame.

I-F07 / I-Q03 | Steel Tube Institute - FE-Based Design of HSS Connections

https://steeltubeinstitute.org/resources/fe-based-design-of-hss-connections/

Connection modeling needs suitable geometry, loading, material representation and verification; elastic and inelastic models address different questions. Connection FE analysis needs appropriate elements, realistic load/support boundaries, local geometry, convergence and validated modeling methods.

Limit: No FE model or validated FE capacity is claimed in Phase3I. CAD tessellation and overlap checks are not stress analysis. No finite-element structural analysis was performed in this phase.

I-B01 | Steel Tube Institute - HSS Beam to HSS Column Connections

https://steeltubeinstitute.org/resources/hss-beam-to-hss-column-connections/

Shear/fixity detailing, local wall load-path checks, angle eccentricity caveat. Building examples are not adopted rail-vehicle approvals.

I-B03 | Steel Tube Institute - Stepped HSS T- and Cross-Connections Under Branch In-Plane and Out-of-Plane Bending

https://steeltubeinstitute.org/resources/stepped-hss-t-and-cross-connections-under-branch-in-plane-and-out-of-plane-bending/

Uneven wall effectiveness, chord plastification/punching and branch wall checks; no direct capacity imported for this vehicle node.

I-B04 | Leeco Steel - ASTM A572 Steel Plate

https://www.leecosteel.com/a572-steel-plate/

Grade 50 minimum Fy 50ksi/Fu 65ksi candidate. Exact quarter-inch stock, heat/CVN, flatness and lead time not quoted.

I-B05 | OnlineMetals - 3 x 3 x 1/4 steel angle, part 9943

https://www.onlinemetals.com/en/buy/carbon-steel/3-x-3-x-0-25-carbon-steel-angle-a36-hot-rolled/pid/9943

Listed 3in legs and 0.25in thickness, cut lengths. Page title says A36 but standards field says ASTM-A529: resolve actual MTR grade before purchase. No procurement approval.

I-B06 | AISC - Fatigue of Welded Connections: A Primer, Part1

https://learning.aisc.org/local/catalog/view/product.php?productid=547

Fatigue depends on stress range, connection geometry and expected life; static stress screening alone cannot release this node.

I-D02 | SSAB ASTM A36 plate

https://www.ssab.com/en-us/brands-and-products/commercial-steel/structural-steel/astm-a36

Finished plate minFy36ksi and Fu58-80ksi; temper-leveled thickness range includes3/16and1/4in.

Limit: No quote or heat certificate. Coil certified chemistry only is not interchangeable with accepted finished plate. Material toughness and fatigue suitability held.

I-D03 | Portland Bolt SAE J429

https://www.portlandbolt.com/technical/specifications/sae-j429/

Grade5 1/4-1in: proof85ksi,yield92ksi,tensile120ksi; J995nuts referenced.

Limit: Material specification comparison, not seat fastener assembly approval, fatigue rating or torque recommendation.

I-D04 | Portland Bolt thread pitch chart

https://www.portlandbolt.com/technical/thread-pitch-chart/

UNC tensileareas:1/2-13 .1419in2;3/8-16 .0775in2.

Limit: Area only; bolt grip, shearplane, threads, coating, locking and nut engagement unresolved.

I-D05 | Norplex-Micarta NP510A technical data bulletin

https://www.norplex-micarta.com/wp-content/uploads/2017/07/TDB_NP510A.pdf

Epoxy-glass FR4. Typical flatwise compression455.1MPa; manufacturer explicitly says typical values are not specification values.

Limit: No allowable assigned. 3mm machined strip, creep, moisture, surface flatness and railfire/smoke qualification require testing.

I-D06 | Plascore Value Add

https://plascore.com/composite-panels/value-add/

Manufacturer offers co-fabricated edge closeouts and inserts.

Limit: No selected insert, pullout/bearing rating, manufacturing tolerance or project floor approval.

I-Q01 | AWS D15.1/D15.1M:2025 public preview

https://pubs.aws.org/Download_PDFS/D15.1-D15.1M-2025-Web_PV.pdf

Public cover, abstract, scope and contents identify the 2025 railroad car/locomotive welding specification, including thin-base-metal provisions.

Limit: Preview is not the full standard; no clause compliance or applicability to this attraction is claimed. Responsible engineer must establish the project welding basis and obtain adopted documents.